

SHAN CAO

Data Scientist

Strategy, consumer insight, and applied AI

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I work where a business question meets the system that can answer it. Eight years across insurance analytics, now in strategy and consumer insight: claims decisions with clinicians, market questions for a beauty company, and the data marts, models, and agent workflows underneath both.

TECHNICAL

- Python, SQL, classical ML, NLP
- Optimization, forecasting, experimentation
- RAG, language-model applications, agent workflows
- Azure, data marts, CI/CD

BUSINESS

- Consumer and market insights
- Insurance claims, compliance, fraud and waste
- KPI design and recommendation campaigns
- Decisions with clinicians and commercial teams

LANGUAGES

- English — professional
- French — intermediate
- Chinese — native

EXPERIENCE

Coty Jun 2026 – Dec 2026

Strategy, CMI & Analytics · Consumer & Market Insights · Paris

- Brought strategy and consumer-insight questions onto one analytics layer: automated consolidation, a data mart, and CI/CD, so market and performance reviews no longer start from scattered sources.
- Placed an agent flow on that layer, turning repeat CMI requests into a workflow the strategy team can run rather than a fresh analysis each time.

Manulife 2022 – 2025

Data Scientist

- Worked beside clinicians to put hospital claims on a straight-through path: retrieval-augmented generation and open-source language models with NLP and classical ML, running end to end on Azure.
- Set fraud, waste, and abuse and reasonable-and-customary thresholds with doctors and the business, joining statistics, optimization, knowledge graphs, and public data to a reimbursement decision.
- Carried the same agent pattern into email, classifying, answering, and routing customer and distributor requests while keeping personal data inside the process.
- Moved 9 projects across 5 markets onto Azure, then used that shared base for 8 personalized recommendation campaigns and a common Asia digital KPI matrix.

Manulife 2019 – 2022

Compliance Data Scientist

- Connected agent-misconduct signals across Southeast Asian markets in one model: a knowledge graph from SQL and Python into Neo4j, then a front end compliance teams used for oversight.
- Read call transcripts with NLP and PCA so simple calls could move to a chatbot, and forecast call volume so planning could follow expected demand.

Qixin Tech 2018 – 2019

AI Lab Researcher

- Turned retail client profiles into data marts and recommendation models, linking what the data showed to who marketing should reach and what they were likely to buy.
- Published the implemented work as first author at IEEE ICAIBD 2019.

EDUCATION

ESSEC & CentraleSupélec 2025 – present

MSc Data Sciences & Business Analytics, AI in healthcare. Business intelligence, decision analysis, deep learning, management. Project: full-stack ML pipeline with CI/CD, Docker, FastAPI, and React. ML fairness and bias.

University of New South Wales 2016 – 2018

Master of Engineering Science, Systems and Control.

Shandong University 2012 – 2014

BEng Detection & Control and BEcon Finance, dual degree.